

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech.

Branch : Electronics Engineering and Allied

Semester :IV

Subject Code & Name: BTETC402/BTEXC402 & Signals and Systems

Max Marks: 60

Date: 21/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The area under the curve $\int_{-\infty}^{\infty} \delta(t) dt$ is	CO1	1
	a. ∞ b. Unity c. Zero d. Undefined		
2	Periodic Signals are	CO1	1
	a. $x(t+T)=x(t)$ b. $x(t-T)=x(t)$ c. $x(t+kT)=x(t)$ d. All of the above		
3	Unit step response of an LTI system with impulse response $h(n) = \delta(n) - \delta(n-1)$ is	CO2	1
	a. $\delta(n-1)$ b. $\delta(n)$ c. $u(n-1)$ d. $u(n)$		
4	The convolution of rectangular pulse with itself is	CO2	1
	a. another rectangular pulse b. square pulse c. triangular pulse d. sinc pulse		
5	The Fourier series of a real, even periodic signal will contain only	CO3	1
	a. cosine terms b. sine terms c. even terms d. odd harmonics		
6	If the Fourier series coefficients of a signal are periodic then the signal must be	CO3	1
	a. continuous-time, periodic b. discrete-time, periodic c. continuous-time, non-periodic d. discrete-time, non-periodic		

7	The Fourier transform of a signal $x(t) = e^t u(-t)$ is given by	CO4	1
	a. $\frac{1}{2-j\omega}$ b. $\frac{1}{1-j\omega}$ c. $\frac{1}{2j-\omega}$ d. $\frac{2}{2j-\omega}$		
8	Fourier transform of a rectangular pulse is	CO4	1
	a. rectangular pulse b. triangular pulse c. sinc pulse d. impulse function		
9	Fourier transform of a real and even signal $x(n)$ is _____ function of frequency	CO4	1
	a. real & even b. real & odd c. complex d. imaginary		
10	ROC of $X(s)$ contain	CO5	1
	a. zeros b. poles c. no zero d. no pole		
11	Laplace transform converts convolution of time signals to	CO5	1
	a. addition b. subtraction c. multiplication d. division		
12	The z-transform of $\delta(n - m)$ is	CO5	1
	a. z^{-n} b. z^{-m} c. $\frac{1}{z^{-n}}$ d. $\frac{1}{z^{-m}}$		
Q. 2	Solve the following.		12
A)	Explain the basic properties of the system.	CO1	6
B)	Draw the following signals:	CO1	6
	i) $u(t-2)$ iii) $u(-t+1)$ v) $u(2t)$		
	ii) $u(-t)$ iv) $u(-t-1)$ vi) $2 \cdot u(t)$		
Q.3	Solve the following.		12
A)	i) Compute the following convolution sum $y(n) = x(n) * h(n)$	CO2	6
	$x(n) = h(n) = u(n)$		
	ii) Compute the following convolution sum $y(n) = x(n) * h(n)$ with the tabular method of convolution sum.		
	$x(n) = \{1, 1, 2, 3\}$, $n=0$ to 3 $h(n) = \{1, 1, 1\}$, $n=0$ to 2		
B)	Derive the formula for convolution integral for continuous time LTI	CO2	6

systems.

Q. 4 Solve Any Two of the following. **12**

A) i) Explain impulse sampling process with neat diagram. **CO1 6**

ii) Find the sampling frequencies f_s for the following signals

a) $x(t) = \sin(200\pi t)$ b) $x(t) = \sin^2(200\pi t)$ c) $x(t) = 1 + \sin(200\pi t)$

B) Find the trigonometric Fourier Series for the following signal **CO3 6**
 $x(t) = 10t/2\pi$, $0 < t < 2\pi$.

C) Derive the formula for Trigonometric Fourier Series for the waveforms **CO3 6**
with even symmetry.

Q.5 Solve Any Two of the following. **12**

A) State and prove any 4 properties of CTFT. **CO4 6**

B) Find the Fourier transform of the sequences: **CO4 6**

i) $x(n) = a^n u(n)$ for $|a| < 1$

ii) $x(n) = a^n u(-n-1)$ for $|a| > 1$

C) State the properties of z-transform. **CO5 6**

Q. 6 Solve Any Two of the following. **12**

A) Determine the Laplace transform and plot it's ROC of the signals: **CO5 6**

i) $x(t) = \sin(w_0 t) u(t)$

ii) $x(t) = \sin(w_0 t) u(t)$

B) State and prove any 4 properties of Laplace transform. **CO5 6**

C) Find the inverse z-transform of $X(z)$ for $|z| > 1/3$ **CO5 6**

$$X(z) = \frac{1 + z^{-1}}{1 - \frac{1}{3}z^{-1}}$$

*** End ***